

Rudra Gohil

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PROJECTS

Monk - A Programming Language From Scratch

- Monk is implemented as a **tree-walking style interpreter**. Tokenization, **Pratt-Parsing** and Evaluation is performed inside the REPL **without using any external Libraries**.
- It also has **functions**, **conditionals**, **implicit and explicit returns**, **recursive functions** and who doesn't love **closures** !

A Raytracer From The Streets

- It was a Mosaic of Programming and Math, Ranging from **Recursion for Reflections**, Object Oriented Programming to glue different Light Sources all the way to realm of **Linear algebra** and **sphere geometry** with numpy.
- Shadow Coherence was taken care of to **increase the efficiency** and Quaternions were used to **implement rotation** and some **multiprocessing** at the end saved the day.

Girgit - Learn Git by Making one

- The Finale of Scratch Trilogy is girgit - a git like version control system **resembling a big linked-list** separating each little nodes with **SHA256**.
- A clear separation was made between the hash object(data) affecting functions, higher level data structures that use those functions and Command line Interface functionalities to ensure tidyness. The crown jewel was native implementation of **Myer's Algorithm** for diffing.
- Also a command with usage similar to **gitk tool** was implemented to **visualize the DAG tree** along with other standard commands.

DinoPDF - a game that lives inside pdf

open this pdf in chrome to play the game...

- The Portable Document Format is widely used to share **static documents** but it also provides **PDF Javascript API** along with formfields (**Acroforms**). A lot of tweaking and creative workarounds resulted in this game.

EDUCATION

Birla Vishvakarma Mahavidyalaya		Jul. 2023 – Jul 2027
<i>Bachelor of Technology in Information Technology</i>	<i>8.86 CPI</i>	Anand, Gujarat
Kendriya Vidyalaya		2019 – 2023
		Anand, Gujarat
• <i>JEE Mains</i>	<i>95.02 %ile</i>	
• <i>Grade 12</i>	<i>92.2 %</i>	
• <i>Grade 10</i>	<i>93.4 %</i>	

SKILLS

Languages : Python , Go(Beginner), MySQL

Tools : Manim, Git, GitHub, WireShark , Math